

Conjecture 00000002510: tensors of bounded rank do not form a
closed set,
and best rank-2 approximation fails by diverging components

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Statement

Definition: Best low-rank approximation of tensors. Conjecture: The existence of best low-rank approximations is governed by closure defects: unlike matrices, the set of bounded-rank tensors is not closed, and the approximation fails via limiting divergence.

This is a known theorem of de Silva and Lim. The arXiv abstract of <https://arxiv.org/abs/math/0607647> says: “we argue that the naive approach to this problem is doomed to failure because, unlike matrices, tensors of order 3 or higher can fail to have best rank- r approximations”. We give a self-contained, machine-checked proof of the statement in the smallest case, real $2 \times 2 \times 2$ tensors and rank 2, together with the matrix contrast and the general closedness criterion.

Reading

We prove the following four claims. Each one is a conjunct of the Lean theorem `C2510.conjecture_2510`.

- (R1) *Unlike matrices.* For all m, n, r , the set of real $m \times n$ matrices of rank $\leq r$ is closed. Hence every real matrix has a best rank- $\leq r$ approximation.
- (R2) *Governed by closure.* For every r : every tensor has a best rank- $\leq r$ approximation if and only if the set of tensors of rank $\leq r$ is closed.
- (R3) *Bounded-rank tensors are not closed.* The set of $2 \times 2 \times 2$ tensors of rank ≤ 2 is not closed. The tensor W below has rank 3 and lies in the closure. Its distance to the rank- ≤ 2 set is 0, but no rank- ≤ 2 tensor attains it, so W has no best rank- ≤ 2 approximation.
- (R4) *Limiting divergence.* Take any sequence of decompositions $S_n = x_n \otimes y_n \otimes z_n + x'_n \otimes y'_n \otimes z'_n$ with $S_n \rightarrow W$. Then the norms of *both* summands tend to $+\infty$.

We read “the set of bounded-rank tensors is not closed” as: for some tensor space and some rank bound r , the set of rank- $\leq r$ tensors is not closed. We do not claim this for every r ; for instance, rank ≤ 0 gives the closed set $\{0\}$. Only the four claims above are proved.

Conventions and definitions

- Tensors are $T : \{0, 1\}^3 \rightarrow \mathbb{R}$ (Lean: `Fin 2 -> Fin 2 -> Fin 2 -> ℝ`). The norm is Mathlib’s default sup norm $\|T\| = \max_{i,j,k} |T_{ijk}|$, with $\text{dist}(S, T) = \|S - T\|$. Closedness, closure and

the divergence statement do not depend on the norm, since all norms are equivalent in finite dimension. “Best approximation” and the infimum are taken for this sup norm.

- $e_1 = (1, 0)$ and $e_2 = (0, 1)$ (indices 0, 1). The outer product is $(a \otimes b \otimes c)_{ijk} = a_i b_j c_k$.
- T has rank $\leq r$ (`HasRankLE T r`) if $T = \sum_{s < r} a_s \otimes b_s \otimes c_s$ for some vectors $a_s, b_s, c_s \in \mathbb{R}^2$. Also `rankLE r` is the set of such T , and `tensorRank T` is the least such r (an `sInf` of natural numbers).
- Matrix rank is Mathlib’s `Matrix.rank`, the dimension of the column space.
- y is a best approximation of x from A (`IsBestApprox A x y`) if $y \in A$ and $\text{dist}(x, y) \leq \text{dist}(x, z)$ for every $z \in A$.
- $W = e_1 \otimes e_1 \otimes e_2 + e_1 \otimes e_2 \otimes e_1 + e_2 \otimes e_1 \otimes e_1$, so $W_{001} = W_{010} = W_{100} = 1$ and all other entries are 0.

Proofs

Lemma 1 (`bestApprox_forall_iff_isClosed`). *Let A be a nonempty subset of a proper metric space X . Then every $x \in X$ has a best approximation from A if and only if A is closed.*

Proof. If A is closed, then Mathlib’s `IsClosed.exists_infDist_eq_dist` gives $y \in A$ with $\text{dist}(x, y) = \inf_{z \in A} \text{dist}(x, z)$. Conversely, let $x \in \overline{A}$ and let y be a best approximation. For each $\varepsilon > 0$ there is $z \in A$ with $\text{dist}(x, z) < \varepsilon$. Hence $\text{dist}(x, y) < \varepsilon$, so $x = y \in A$. $\square$

(R2) follows, because $\mathbb{R}^{2 \times 2 \times 2}$ is proper and $0 \in \text{rankLE } r$.

Theorem 2 ((R1), `matrix_rankLE_isClosed`, `matrix_bestApprox_exists`). $\{M \in \mathbb{R}^{m \times n} : \text{rank } M \leq r\}$ is closed.

Proof. The map $M \mapsto (v \mapsto Mv)$ from matrices to continuous linear maps is linear on a finite-dimensional space, hence continuous. The rank of M equals the rank of this linear map. Mathlib’s `isOpen_setOfPred_nat_le_rank` says that $\{f : r+1 \leq \text{rank } f\}$ is open. Our set is the complement of its preimage. The existence of best approximations then follows from the Lemma. $\square$

Lemma 3 (`W_eqns_inconsistent`, `W_not_rankLE_two`). W is not a sum of two rank-one tensors.

Proof. Suppose $W = a \otimes b \otimes c + p \otimes q \otimes r$. This gives eight equations $a_i b_j c_k + p_i q_j r_k = W_{ijk}$. Put $D = a_0 p_1 - a_1 p_0$. Take p_1 times the $(0, j, k)$ equation minus p_0 times the $(1, j, k)$ equation (Cramer’s rule on the first-mode slices). This gives $Db_j c_k = p_1 W_{0jk} - p_0 W_{1jk}$, that is, $Db_0 c_0 = -p_0$, $Db_0 c_1 = Db_1 c_0 = p_1$ and $Db_1 c_1 = 0$. Since $(Db_0 c_1)(Db_1 c_0) - (Db_0 c_0)(Db_1 c_1) = 0$ identically, we get $p_1^2 = 0$. In the same way, a_0 times the $(1, j, k)$ equation minus a_1 times the $(0, j, k)$ equation gives $Dq_0 r_0 = a_0$, $Dq_0 r_1 = Dq_1 r_0 = -a_1$ and $Dq_1 r_1 = 0$, so $a_1^2 = 0$. Then the $(1, 0, 0)$ equation $a_1 b_0 c_0 + p_1 q_0 r_0 = 1$ reads $0 = 1$. Every step is a `linear_combination` in Lean. $\square$

Hence W is not of rank $\leq 0, 1, 2$, using the padding lemma `HasRankLE.mono`. The three-term definition of W is a rank-3 decomposition (`W_hasRankLE_three`), so `tensorRank W = 3` (`tensorRank_W`).

Theorem 4 ((R3), `Tseq_mem`, `Tseq_tendsto`, `W_mem_closure`, `rankLE_two_not_isClosed`, `infDist_W`, `W_no_best_approx`). Let $t_n = 1/(n+1)$ and $u_n = e_1 + t_n e_2$. Then $T_n = (n+1)u_n \otimes u_n \otimes u_n - (n+1)e_1 \otimes e_1 \otimes e_1$ has $\text{rank} \leq 2$ and $T_n \rightarrow W$. Consequently `rankLE2` is not closed, $\inf_{S \in \text{rankLE2}} \|W - S\| = 0$, and W has no best $\text{rank} \leq 2$ approximation.

Proof. An entry-by-entry check (`Tseq_eq`) gives $T_n = W + t_n P + t_n^2 Q$, where $P = e_1 \otimes e_2 \otimes e_2 + e_2 \otimes e_1 \otimes e_2 + e_2 \otimes e_2 \otimes e_1$ and $Q = e_2 \otimes e_2 \otimes e_2$. Since $t_n \rightarrow 0$, we get $T_n \rightarrow W$, so W lies in the closure. But $W \notin \text{rankLE2}$ by the Lemma, so the set is not closed. The infimum is 0 by `Metric.mem_closure_iff_infDist_zero`. A best approximation S would satisfy $\|W - S\| \leq \|W - T_n\| \rightarrow 0$, so $W = S \in \text{rankLE2}$, which is impossible. $\square$

Theorem 5 ((R4), `max_summand_tendsto`, `summand_tendsto`). Let x_n, y_n, z_n (summand 0) and x'_n, y'_n, z'_n (summand 1) be vectors in $\mathbb{R}^2$ with $S_n = x_n \otimes y_n \otimes z_n + x'_n \otimes y'_n \otimes z'_n \rightarrow W$. Then $\|x_n \otimes y_n \otimes z_n\| \rightarrow \infty$ and $\|x'_n \otimes y'_n \otimes z'_n\| \rightarrow \infty$.

Proof. First, $\max(\|\text{summand 0}\|, \|\text{summand 1}\|) \rightarrow \infty$. Otherwise there are C and a subsequence φ on which both summand norms are $< C$. In the sup norm, $\|a\| \|b\| \|c\| \leq \|a \otimes b \otimes c\|$ (`norm_mul_le_norm_outer3`). So each summand can be rewritten with factors $(\|b\| \|c\|)a$, $b/\|b\|$, $c/\|c\|$ (or 0, 0, 0 if $b = 0$ or $c = 0$), all of norm $\leq \max(C, 1)$ (`rebalance`). By Bolzano–Weierstrass (`Mathlib's tendsto_subseq_of_bounded`), a further subsequence of these factor pairs converges. By continuity of $(a, b, c) \mapsto a \otimes b \otimes c$, the limit is a two-term decomposition of $\lim S_{\varphi(\psi(n))} = W$. This contradicts the Lemma. Second, the triangle inequality gives $\|\text{summand } s\| \geq \max(\cdot, \cdot) - \|S_n\|$, and eventually $\|S_n\| < \|W\| + 1$. $\square$

The Wikipedia article https://en.wikipedia.org/wiki/Tensor_rank_decomposition describes this phenomenon (the summand norms tending to infinity) and says: “It is sometimes called the problem of diverging components.”

Lean

The file `lean/Conjecture2510/Basic.lean` (namespace `C2510`, only `import Mathlib`) proves the main theorem `C2510.conjecture_2510`. It is the conjunction of (R1) for all m, n, r , (R2) for all r , non-closedness of `rankLE2`, $W \in \overline{\text{rankLE2}}$, $\text{tensorRank } W = 3$, $\text{infDist}(W, \text{rankLE2}) = 0$, the absence of a best approximation, and (R4) for both summands. `lake build` succeeds, and `#print axioms C2510.conjecture_2510` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.